

§8 Digital Addiction

1.1 Research question(s) addressed in this topic

本课题研究数字技术使用中的“数字成瘾”问题，重点是高频使用社交媒体和智能手机是否完全反映真实消费效用，还是部分来自习惯形成和自我控制问题。具体而言，研究关心短期减少高诱惑 app 使用是否会改变后续使用习惯，以及人们是否愿意主动使用承诺装置来限制未来的自己。其更广泛的问题是：数字产品的高使用量在多大程度上代表消费者福利，在多大程度上反映行为偏差。

This topic studies “digital addiction” in the use of digital technologies. It asks whether heavy use of social media and smartphones fully reflects genuine consumer surplus, or whether it is partly driven by habit formation and self-control problems. More specifically, it examines whether temporarily reducing the use of tempting apps changes later habits, and whether people voluntarily demand commitment devices to restrict their future behavior. The broader issue is whether high digital consumption should be interpreted as welfare-enhancing consumption or as behavior partly shaped by biases.

1.2 Summarize the design and purpose of the set of experiments in this topic

实验采用随机田野实验。研究者招募手机用户，让他们安装一个专门设计的手机追踪工具，用来记录真实屏幕使用时间，并在实验过程中多次收集调查数据。研究对象集中在若干类高诱惑 app，同时收集实际使用、未来使用预测、主观幸福感和成瘾相关问卷指标。

实验主要包括两个处理。第一是 *bonus treatment*，即在特定三周内为减少高诱惑 app 使用提供临时金钱激励，用来研究短期激励和习惯形成之间的关系。第二是 *limit treatment*，即向参与者提供未来生效的屏幕时间限制功能，并使用 MPL，Multiple Price List，估计他们为了获得该功能愿意放弃的现金数量。这个设计用于考察人们是否愿意主动牺牲金钱来约束未来的自己，从而检验自我控制问题。

The study uses a randomized field experiment. Participants were phone users who installed a specially designed tracking tool that recorded real screen time of several tempting apps. Related surveys are done to collect additional information such as subjective well-being etc.

There were two main treatments. The first was the bonus treatment, which provided temporary monetary incentives for reducing the use of tempting apps during a specific three-week period. Its purpose was to study the relation between short-run incentives and habit formation. The second was the limit treatment, which gave participants access to future-effective screen-time limits. The researchers also used a Multiple Price List to estimate how much cash participants were willing to give up to obtain this function. This design was used to examine whether people voluntarily sacrifice money to restrict their future selves, thereby testing self-control problems.

1.3 Summarize the main experimental findings in this topic

主要发现是，临时金钱激励降低了高诱惑 app 使用，并且这种影响在激励结束后仍然存在。这表明数字产品使用具有习惯形成特征：短期减少使用不仅是对当期激励的反应，也会影响后续使用倾向。

另一个主要发现是，屏幕时间限制功能也降低了高诱惑 app 使用。许多参与者主动设置限制，并且在没有额外奖励的情况下愿意放弃一定现金来获得限制功能。这说明人们意识到自己未来可能会过度使用，因此愿意购买一种约束未来行为的承诺装置。结合结构模型，作者认为自我控制问题解释了社交媒体使用中的重要部分。

The main finding is that temporary monetary incentives reduced the use of tempting apps, and the effect persisted after the incentives ended. This suggests habit formation: short-run reductions in use are not merely responses to current incentives, but also affect later demand.

Another key finding is that screen-time limits also reduced the use of tempting apps. Many participants voluntarily set limits, and they were willing to give up some cash to obtain access to the limiting function even without additional rewards for using it. This suggests that people recognize possible future overuse and value a commitment device. Combined with the structural model, the authors argue that self-control problems explain an important part of social media use.

1.4 Critical thoughts on the experimental design and the authors' interpretation

第一，从内部有效性看，实验中的测量和调查过程本身可能改变参与者行为。参与者知道自己的手机使用被记录，并且需要反复完成调查、预测未来使用、回答和手机成瘾相关的问题。这些过程可能提高他们对屏幕时间的注意力，使他们有意识地减少使用。因此，使用量下降未必完全来自习惯形成或承诺装置，也可能部分来自被持续提醒和被研究情境影响。

第二，从解释范围看，实验主要识别的是手机上高诱惑 app 使用的变化，但不能完全排除替代效应。参与者可

能转向其他 app，或者改用电脑、平板等未被同样精确记录的设备。因此，将结果解释为总体数字消费减少或福利必然提高需要谨慎。该实验更有力地说明特定手机 app 使用下降，而不是完整证明数字成瘾的全部福利后果。

First, regarding internal validity, the measurement and survey process itself may have changed participants' behavior. Participants knew that their phone use was being recorded and repeatedly answered questions about future use and phone addiction. These procedures may have increased their attention to screen time and made them consciously reduce use. Therefore, the observed reduction may not be driven only by habit formation or commitment demand, but also by repeated reminders and the experimental context.

Second, regarding the scope of interpretation, the experiment mainly identifies changes in the use of tempting apps on smartphones, but it cannot fully rule out substitution. Participants may switch to other apps, or to computers and tablets that are not measured with the same precision. Therefore, it is safer to interpret the results as evidence of reduced use of specific smartphone apps, rather than as complete evidence that total digital consumption falls or welfare necessarily increases.

§9 Status Goods

1.1 Research question(s) addressed in this topic

本主题研究消费者是否会为商品的可见身份信号付费，即购买白金信用卡是否不仅因为额度、服务和折扣，也因为它能让他人推断自己收入较高。进一步的问题是，身份商品是否存在位置外部性，以及社会形象需求是否部分来自心理层面的自我形象补偿。

This topic asks whether consumers pay for the visible status signal of a good. In the platinum-card setting, the question is whether demand reflects only functional benefits or also the desire to be perceived as high-income. It also studies whether status goods create positional externalities and whether social image partly substitutes for self-image.

1.2 Summarize the design and purpose of the set of experiments in this topic

第一组实验向金卡客户提供真实付费升级机会。处理组获得真正白金卡，对照组获得相同权益但普通外观的卡，测量是否接受升级，用来检验可见身份标签的价值。第二组实验向白金卡客户提供付费钻石卡升级，并操纵他们是否得知白金卡门槛降低，用来检验排他性的作用。第三组实验通过自我肯定任务操纵自尊，并测量对白金卡或奢侈品牌礼品卡的需求，用来检验自我形象机制。

The first experiment offered gold-card customers a real paid upgrade. The treatment group received an actual platinum card, while the control group received identical benefits on a non-platinum-looking card. The outcome was upgrade take-up. The second experiment offered platinum-card holders a paid diamond-card upgrade and varied information about lower access requirements for platinum cards. The final experiments manipulated self-esteem and measured demand for status goods.

1.3 Summarize the main experimental findings in this topic

主要发现是，真正白金卡的需求高于相同功能但无白金外观的产品，说明可见身份信号本身有价值。交易数据也显示，白金卡更常用于餐厅、酒吧等社交场景。相对低收入客户更重视这种身份信号。白金卡门槛降低的信息提高了钻石卡需求，说明身份商品存在位置外部性。提高自尊会降低对身份商品的需求，作者据此认为社会形象和自我形象更像替代关系。

The main finding is that demand for the actual platinum card exceeds demand for the same benefits without the visible platinum appearance. This suggests that the status signal itself has value. Transaction evidence shows that platinum cards are used more in socially visible settings. Demand is stronger among relatively lower-income customers. Information that weakens platinum exclusivity increases demand for the diamond card. Higher self-esteem reduces demand for status goods, suggesting that social image and self-image are substitutes.

1.4 Critical thoughts on the experimental design and the authors' interpretation

第一个批判点是内部有效性。真正白金卡可能带来未被完全控制的实际便利，例如商户、银行人员或客服看到卡面后给予不同识别或待遇。因此，接受率差异可能不仅反映社会形象，也可能包含对服务质量或机构认可的预期。

第二个批判点是福利解释。实验说明消费者愿意为身份信号付费，也说明排他性重要，但不能直接证明身份消费是浪费。身份信号可能带来商业信任、社交连接或真实心理效用；要得出福利损失结论，还需要额外证据。

One internal-validity concern is that the actual platinum card may provide practical advantages not fully controlled by the experiment, such as better recognition by merchants, bank staff, or customer service. Thus, the take-up gap may partly reflect expected service quality, not only social image.

A second concern is welfare interpretation. The experiments show demand for status and exclusivity, but they do not

directly prove that status consumption is wasteful. Status signals may generate trust, social connections, or genuine psychological utility.

§10 Peer Effects

1.1 Research question(s) addressed in this topic

本主题研究同伴是否会影响个人决策，以及如何把真正的同伴影响与共同背景、共同环境和自我选择区分开。教育部分关注大学生的学习结果是否受同组同学能力构成影响，尤其是能力分组能否改善低能力和中等能力学生的表现。金融部分关注朋友的投资选择为什么会影响个人购买金融资产，并进一步区分这种影响来自学习朋友的购买信息，还是来自朋友实际持有同一资产所带来的社会效用。

This topic asks whether peers causally affect individual decisions and how such effects can be separated from common background, shared environments, and selection. In the education setting, it studies whether the ability composition of tutorial groups affects students' academic outcomes, especially whether ability grouping benefits low- and middle-ability students. In the financial setting, it asks why a peer's investment choice affects one's own willingness to buy an asset, distinguishing learning from the peer's decision from the social utility of the peer actually owning the same asset.

1.2 Summarize the design and purpose of the set of experiments in this topic

教育实验将大学一年级学生按入学前 GPA 类别随机分配到不同能力构成的 tutorial groups。关键操纵是同伴平均能力和能力离散程度，主要测量学分、成绩和退学情况，用来检验小组能力构成如何影响学生表现，并比较混合分组、两层分组和三层分组等方案。金融实验识别有社会关系的投资者对，先让 investor 1 面对真实资产购买机会，再用彩票决定其是否买到，并随机告知 investor 2 不同信息，用来分离社会学习和社会效用。

The education experiment randomly assigned first-year students, conditional on prior GPA categories, to tutorial groups with different ability compositions. The key manipulated variables were peers' mean ability and dispersion, and the main outcomes were credits, grades, and dropout. This design was used to test how peer ability composition affects achievement and to compare alternative grouping rules. The financial experiment paired socially connected investors. Investor 1 first faced a real investment opportunity; a lottery determined whether he obtained the asset; investor 2 then received randomized information. This design separated social learning from social utility.

1.3 Summarize the main experimental findings in this topic

教育实验发现，同伴效应具有非线性和异质性。学生表现通常随同伴平均能力提高而改善，但同伴能力差异过大可能不利。三层能力分组主要提高低能力和中等能力学生的学业结果，并降低退学风险，高能力学生基本不受影响。金融实验发现，朋友想买资产会提高 investor 2 的购买意愿，说明社会学习重要；朋友实际获得资产还会进一步提高购买意愿，说明共同持有本身也有额外效用。金融知识较弱者更受朋友信息影响。

The education experiment finds nonlinear and heterogeneous peer effects. Student performance tends to improve with higher mean peer ability, while excessive dispersion in peer ability can be harmful. Three-way tracking mainly benefits low- and middle-ability students and reduces dropout risk, while high-ability students are largely unaffected. The financial experiment finds that knowing a peer wanted to buy raises investor 2's purchase probability, supporting social learning. Knowing that the peer actually received the asset raises it further, suggesting an additional social-utility channel. The learning effect is stronger among less financially sophisticated investors.

1.4 Critical thoughts on the experimental design and the authors' interpretation

能力分组的效果可能混合了同伴效应和标签效应。学生若能意识到自己所在小组的相对能力位置，分到低、中、高能力组本身就可能改变信心、自我评价和努力程度。因此，成绩和退学变化未必完全来自同伴互动或能力构成。

金融实验中，“朋友想买但没买到”可能同时传递资产稀缺、额度有限或需求较强的信息，因此 B-A 不一定是纯粹的社会学习效应。并且，信息由经纪人在销售电话中传递，investor 2 也可能将其理解为营销话术。

The effect of ability grouping may mix peer effects with labeling effects. If students infer the relative ability level of their assigned group, placement itself may affect confidence, self-perception, and effort. Thus, changes in grades or dropout may not be driven only by peer interaction or ability composition.

In the financial experiment, “the peer wanted to buy but did not receive the asset” may also signal scarcity, limited access, or strong demand. Hence, B-A may not capture pure social learning. Because the information is delivered by a broker during a sales call, investor 2 may also interpret it as a marketing message.

§11 Policy Uncertainty

1.1 Research question(s) addressed in this topic

本主题研究个人如何面对未来社会保障福利的不确定性，以及这种不确定性会造成多大的主观福利成本。核心问题不是预测未来政策实际会怎样变化，而是测量人们对未来福利的主观分布，并估计他们愿意放弃多少预期福利来换取确定性。更一般地说，它关注没有市场价格的政策风险如何被转化为可测量的风险溢价。

This topic asks how individuals perceive uncertainty about future Social Security benefits and how costly this uncertainty is for their welfare. The main goal is not to forecast actual policy reform, but to measure people's subjective beliefs about future benefits and estimate how much expected benefit they are willing to give up for certainty. More broadly, it studies how policy risk without a market price can be translated into a measurable risk premium.

1.2 Summarize the design and purpose of the set of experiments in this topic

研究者设计了一项互联网调查。受访者先以当前法律下应得福利为基准，用分配小球的方式报告未来社保福利可能落入不同区间的概率，由此得到他们的主观预期分布。随后，受访者在确定福利合同和自己报告的不确定福利分布之间进行一系列选择，用来估计确定性等价。研究者还随机改变示例分布、起始值和选项顺序，用来检查回答是否受到问卷框架影响。

The researchers conduct an internet-based survey. Respondents first report a subjective probability distribution of future Social Security benefits relative to benefits under current law, using a visual allocation task. They then make a sequence of choices between a guaranteed benefit contract and their own uncertain benefit distribution, which is used to elicit a certainty equivalent. The survey also randomizes examples, starting values, and option order to examine whether responses are affected by the survey format.

1.3 Summarize the main experimental findings in this topic

主要发现是，受访者普遍认为未来社保福利存在明显不确定性，并且其确定性等价低于主观预期福利。这说明政策不确定性本身带来正的风险溢价，人们愿意放弃一部分预期福利来消除这种不确定性。作者还发现，这种风险溢价在更接近退休、调整空间较小，或经济缓冲能力较弱的人群中更高。因此，政策不确定性不仅影响宏观环境，也会直接进入个体福利。

The main finding is that respondents perceive substantial uncertainty about future Social Security benefits, and their certainty equivalents are below their subjective expected benefits. This implies a positive risk premium for policy uncertainty: people are willing to give up part of their expected benefits to remove uncertainty. The estimated cost is higher for individuals with less time or fewer resources to adjust. Thus, policy uncertainty affects not only aggregate outcomes but also individual welfare directly.

1.4 Critical thoughts on the experimental design and the authors' interpretation

一个内部有效性质疑是，调查任务本身较复杂。受访者需要理解主观概率分布、预期福利和确定性等价，并将这些概念用于假设性的社保合同选择，因此部分回答可能反映理解能力或临时构造的判断，而不完全是稳定偏好。另一个独立问题是确定性等价存在起始值效应。这不能直接证明受访者不理解问题，但说明答案会受到问卷锚点影响。因此，估计的风险溢价应被解释为带有测量误差和框架影响的感知福利成本，而不是完全干净的偏好参数。

A key internal validity concern is that the task is cognitively demanding. Respondents must understand subjective probability distributions, expected benefits, and certainty equivalents, and then apply these concepts to a hypothetical Social Security contract. Some answers may therefore reflect comprehension or constructed judgments rather than stable preferences. A separate issue is the starting-value effect in the certainty-equivalent questions. This does not directly prove misunderstanding, but it shows that reported valuations are sensitive to survey anchors. The estimated risk premium should therefore be interpreted as a perceived welfare cost affected by measurement error and framing, rather than as a fully clean preference parameter.

§12 Asset Bubbles

1.1 Research question(s) addressed in this topic

本主题研究资产泡沫能否在基本价值可控的实验环境中被识别，以及泡沫背后的行为机制。核心问题包括：无经验交易者是否更容易推动价格偏离基本价值；当有经验和无经验交易者混合时，泡沫是否会重新出现；以及在资

产本身没有价值的情况下，交易者是否会因为不确定自己在交易顺序中的位置而进行投机性买入。

This topic studies whether asset bubbles can be identified when fundamental value is experimentally controlled, and what behavioral mechanisms generate them. The central questions are whether inexperienced traders drive prices away from fundamentals, whether bubbles return in markets with both experienced and inexperienced traders, and whether traders speculate on a valueless asset when they are uncertain about their position in a resale sequence.

1.2 Summarize the design and purpose of the set of experiments in this topic

第一组实验构造一个随机股息资产市场。资产存续十期，每期先交易再支付随机股息，剩余期数决定基本价值。市场采用双向拍卖，每组有六名交易者。前几轮保持同组交易者以形成经验，最后一轮替换部分交易者，形成经验混合市场。研究者用价格相对基本价值的偏离来衡量泡沫，并考察交易经验的作用。

第二组实验设置一个序贯转售游戏。资产基本价值为零，三名交易者被随机分配到交易顺序中，但不知道自己的位置。价格外生递增，处理组改变价格上限；研究者记录不同价格和位置信息下的买入选择，用来检验位置不确定性、反向推理能力和对他人接盘行为的预期。

The first experiment creates a dividend-paying asset market. The asset lasts ten periods, trade occurs before dividends are paid, and the remaining periods determine fundamental value. Six traders interact in a double-auction market. The same group trades repeatedly to generate experience, and later some traders are replaced to create mixed-experience markets. Prices relative to fundamentals are used to measure bubbles and study the role of experience.

The second experiment uses a sequential resale game. The asset has zero fundamental value. Three traders are randomly placed in a sequence but do not know their position. Prices rise mechanically, and treatments vary the maximum possible price. Buying choices are used to examine position uncertainty, backward reasoning, and beliefs about later traders' willingness to buy.

1.3 Summarize the main experimental findings in this topic

主要发现是，重复交易经验会降低价格对基本价值的偏离。在传统资产市场中，随着交易者积累经验，泡沫趋于减弱；当无经验交易者后来进入有经验交易者的市场时，典型的泡沫崩盘模式没有明显重新出现。这说明在该实验环境中，有经验交易者能够在一定程度上稳定价格。

在序贯转售游戏中，即使资产没有基本价值，实验中仍然出现投机性买入。有限价格上限下，理论上可以通过从最后一位向前推理排除泡沫，但实际买入仍然存在。买入倾向随离最高价格越远而上升，说明位置不确定性、对他人错误的预期和粗略推理会放大投机行为。

The main finding is that repeated trading experience reduces deviations from fundamental value. In the standard asset market, bubbles become weaker as traders gain experience. When inexperienced traders later enter markets with experienced traders, the typical bubble-crash pattern does not clearly return. This suggests that experienced traders can partly discipline prices in this environment.

In the sequential resale game, speculative buying appears even though the asset has no fundamental value. With finite price caps, backward reasoning from the last trader should rule out bubbles, but buying still occurs. The propensity to buy is higher when the offered price is farther from the maximum price, suggesting that position uncertainty, expected mistakes by others, and coarse reasoning amplify speculation.

1.4 Critical thoughts on the experimental design and the authors' interpretation

一个内部有效性质疑是，传统资产市场中的泡沫可能混入理解不足。资产每期支付随机股息，且基本价值随剩余期数下降，若被试没有完全理解这一动态计算，价格偏离可能部分反映规则理解或计算能力，而不只是投机泡沫。

另一个问题是，序贯转售游戏中的有限责任和外部融资可能人为降低投机成本。交易者最大损失被限制，额外资金由外部融资者提供，这会放大买入意愿，使结果不完全来自位置不确定性或对他人接盘的预期。

One internal-validity concern is that bubbles in the dividend asset market may partly reflect misunderstanding. Since the asset pays random dividends and its fundamental value declines over time, some subjects may not fully compute the changing value. Mispricing may therefore capture limited comprehension or calculation errors, not only speculative trading.

A second concern is that limited liability and outside financing in the resale game reduce the private cost of speculation. Traders' losses are capped, and extra funds are provided by outside financiers. This may inflate buying behavior, so the results may not come only from position uncertainty or beliefs about later resale.

§13 Risk-Taking

1.1 Describe the research question(s) addressed in this topic.

本主题研究的是：为什么父母和子女在股票市场参与、股票持有比例和投资组合风险上存在相似性？这种代际相关究竟来自出生前因素，例如遗传或胎内环境，还是来自出生后的家庭环境，例如学习、模仿和金融信息传递？这个问题重要，因为风险资产长期收益较高，若风险承担行为在家庭中持续传递，可能影响财富积累和财富不平等。

This topic asks why risky financial behavior is correlated across generations. The key question is whether children resemble their parents because of prebirth factors, such as genetic predisposition, or because they learn from parents after birth. This matters because participation in risky financial markets can affect long-run wealth accumulation and therefore may contribute to inequality across families.

1.2 Summarize the empirical strategy in this topic.

研究者利用瑞典收养数据，因为这些数据能同时观察被收养子女的生物父母和养父母。他们考察三个结果：是否持有股票、参与股票市场后的股票投资比例、以及参与后的投资组合波动率。基本思路是比较被收养子女与生物父母、养父母投资行为的相关性。在近似随机分配到养父母家庭的假设下，生物父母的系数反映出生前因素，养父母的系数反映出生后家庭环境。

The empirical strategy uses Swedish adoption data that link adoptees to both their biological and adoptive parents. The authors study equity market participation, the share of financial wealth invested in equities conditional on participation, and portfolio volatility conditional on participation. If adoptees are approximately randomly assigned to adoptive families, correlations with biological parents capture prebirth influences, while correlations with adoptive parents capture postbirth environmental influences.

1.3 Summarize the main empirical results and the main conclusions in this topic.

主要结果是，亲生家庭中的代际相关很强。对被收养子女而言，股票市场参与同时与生物父母和养父母相关，但养父母的影响更大。股票投资比例主要与养父母相关，几乎看不到生物父母影响。投资组合波动率则同时显示一定生物因素和环境因素。总体结论是，出生前因素和出生后环境都重要，但金融风险承担很大程度上由后天家庭环境塑造。

The main results show strong intergenerational persistence in financial behavior. For adoptees, equity market participation is related to both biological and adoptive parents, but the adoptive-parent effect is larger. The risky share is mainly related to adoptive parents, while portfolio volatility shows evidence of both biological and environmental influences. The broad conclusion is that both nature and nurture matter, but postbirth family environment is generally more important for financial risk-taking.

1.4 Do you have any critical thoughts on the authors' interpretation of their results? If so, specify them.

一个问题是，父母投资行为对子女投资行为的影响未必只反映风险偏好或金融态度的传递。投资股票的父母可能拥有更多可转移资源，并通过赠与、继承预期或家庭支持提高子女财富，从而让子女更容易进入股票市场或承担更高风险。另一个问题是，父母和子女投资组合相似也可能来自共同的金融渠道，例如使用相同银行、金融顾问或家庭账户管理方式。因此，结果更稳妥地说明家庭环境会影响金融行为，但不一定能完全说明这种影响来自偏好本身。

One concern is that parental investment behavior may affect children's portfolio choices through wealth rather than only through preferences or attitudes toward risk. Parents who invest more may also provide gifts, expected inheritances, or other family resources that make stock market participation easier. Another concern is that similarity in portfolios may reflect shared financial channels, such as the same bank, adviser, or household account-management practices. Thus, the results support the importance of family environment, but they do not fully pin down whether the channel is preference transmission itself.

§14 Salient Risks

1.1 Describe the research question(s) addressed in this topic.

本主题研究公司管理者在评估风险时是否会受到显著事件影响，从而产生系统性判断偏差。具体而言，当附近发生飓风但公司本身没有受到直接损害时，管理者是否会因为流动性风险变得更容易被想到，而暂时提高对风险的主观判断，并改变公司的现金持有政策。更广义的问题是，真实公司决策中的管理者是否像普通个体一样会使用心理捷径，而这种判断错误是否会影响公司金融政策。

This topic asks whether managers make predictable mistakes in risk assessment when a risk becomes salient. The specific question is whether firms that narrowly avoid a hurricane increase cash holdings because managers temporarily perceive

liquidity risk to be higher, even though the objective probability of being hit does not rise. More broadly, the topic studies whether psychological shortcuts in probability assessment can affect real corporate policies, rather than only laboratory choices or survey responses.

1.2 Summarize the empirical strategy in this topic.

研究者利用飓风作为外生且显著的风险事件。核心比较对象是位于灾区附近但没有被直接击中的公司，以及距离灾区较远的公司。前者经历了“差点被击中”的风险提醒，但没有真实灾害损失；后者作为对照组。研究者用公司现金持有占资产比例衡量管理者对流动性风险的反应，并加入公司季节固定效应、时间固定效应和公司控制变量。研究还考察公司文件中是否更多提到飓风风险，用来辅助判断现金变化是否来自风险感知变化。

The empirical strategy uses hurricanes as salient but plausibly exogenous events. The treated firms are those located near the disaster area but not directly hit, while control firms are farther away. This comparison is designed to isolate a change in perceived risk without a direct change in firm fundamentals. Cash holdings relative to assets are used as a behavioral measure of perceived liquidity risk. The regressions include firm-season fixed effects, time fixed effects, and firm controls. The study also examines whether firms mention hurricane risk more often in regulatory filings, which helps connect cash policy to risk perception.

1.3 Summarize the main empirical results and the main conclusions in this topic.

主要结果是，未被飓风直接击中的邻近公司会显著增加现金持有，幅度约为总资产的 1.1 个百分点。这个增加是暂时的，主要出现在飓风后的几个季度，随后回到原有水平。类似反应主要发生在第一次和第二次附近飓风之后，重复出现后明显减弱。同时，邻近公司也更可能在公开文件中提到飓风风险。结论是，管理者会对显著风险过度反应，流动性风险被短暂高估，并转化为真实的公司现金政策调整。

The main finding is that firms near the disaster area, but not directly affected, increase cash holdings by about 1.1 percentage points of assets. The effect is temporary: cash rises after the hurricane and later reverts to its previous level. The response is strongest after the first and second near-miss events and fades after repeated exposure. These firms also mention hurricane risk more often in filings. The conclusion is that managers overreact to salient risks: a vivid nearby disaster raises perceived liquidity risk and leads to precautionary cash accumulation, even though objective risk does not appear to increase.

1.4 Do you have any critical thoughts on the authors' interpretation of their results? If so, specify them.

可以提出几点谨慎看法。第一，邻近公司虽然没有被直接击中，但可能仍受到供应链、客户需求、本地融资环境或保险市场变化的间接影响，因此现金增加未必完全来自心理偏差。第二，公司总部位置不一定准确反映实际经营和资产暴露，若公司在灾区附近有工厂、客户或供应商，真实风险可能被低估。

Several cautions are possible. First, firms near a hurricane may not be directly hit, but they could still face indirect changes in supply chains, customer demand, local credit conditions, or insurance markets. Thus, the cash increase may not be purely psychological. Second, headquarters location may not fully capture a firm's real exposure if operations, suppliers, or customers are located near the disaster area.

§15 Confirmatory Bias

1.1 Describe the research question(s) addressed in this topic.

本主题研究金融市场参与者是否会受到确认性偏误影响，即人们是否更容易接受支持既有看法的信息，而低估或忽略与原有看法相冲突的信息。进一步的问题是，这种有偏的信息处理是否会影响信念更新、分析师分歧和金融市场中的信息反应。

This topic asks whether financial market participants process information in a biased way. In particular, it studies whether people react less to information that contradicts their prior beliefs. It also asks whether such biased updating can generate disagreement and affect how financial information is incorporated into market expectations.

1.2 Summarize the empirical strategy in this topic.

研究者先建立一个简单模型，说明当部分交易者忽略与先前信念相冲突的信息时，会产生可检验的预测。核心预测是：当过去信息方向不一致时，人们对新信息的修正更弱，信念分歧更大。

实证上，研究者使用美国公司季度盈余公告和分析师盈余预测数据。盈余公告中的好消息或坏消息被用来代表新信息，分析师预测修正被用来衡量信念更新，分析师预测之间的离散程度被用来衡量分歧。

The authors first develop a simple model in which some traders underreact to information that conflicts with their prior beliefs. The model predicts weaker belief revision and larger disagreement when recent signals have different directions.

Empirically, they use US firms' quarterly earnings announcements and analysts' earnings forecasts. Earnings news is used as a measure of new information, forecast revisions measure belief updating, and forecast dispersion measures disagreement across analysts.

1.3 Summarize the main empirical results and the main conclusions in this topic.

主要结果支持确认性偏误的解释。分析师在看到好消息后，如果之前出现过坏消息，就更不容易上调预测；在看到坏消息后，如果之前出现过好消息，也更不容易下调预测。分析师自己的先前预测方向也会影响其后续反应。

此外，当过去几个季度盈余消息方向不一致时，分析师之间的预测分歧更大。总体结论是，金融市场中的信息处理并不总是完全理性的，既有信念会影响人们如何解释和吸收新信息。

The main findings are consistent with confirmatory bias. Analysts are less likely to revise forecasts upward after good news if earlier news was negative, and less likely to revise downward after bad news if earlier news was positive. Their own previous forecast direction also affects later updating.

The authors also find that forecast dispersion is larger when recent earnings news has mixed signs. The main conclusion is that information processing in financial markets is shaped by prior beliefs, rather than being fully neutral and symmetric.

1.4 Do you have any critical thoughts on the authors' interpretation of their results? If so, specify them.

一个可能的质疑是，盈余信息本身的质量并不相同。盈余惊喜只是公开信息的一部分，分析师还可能观察到管理层指引、电话会议内容、行业新闻和一次性会计项目。因此，分析师没有立即顺着盈余惊喜修正预测，不一定说明他们忽略了反向信息，也可能是认为该信息噪声较大。

另一个问题是，预测分歧扩大不一定只来自确认性偏误。连续几个季度盈余消息方向不同，本身可能说明公司经营更不稳定、基本面更难判断。在这种情况下，更大的分析师分歧也可能是对真实不确定性上升的理性反应。

One concern is that earnings information may differ in quality. Earnings surprises capture only part of the available public information. Analysts may also observe management guidance, conference-call discussions, industry news, and one-time accounting items. Therefore, weak forecast revision does not necessarily mean that analysts ignore contradictory information.

A second concern is that larger forecast dispersion may not only reflect confirmatory bias. Mixed earnings news across quarters may indicate genuinely unstable fundamentals. In that case, greater disagreement among analysts could be a rational response to higher uncertainty, rather than purely a psychological bias.

§16 Social Responsibility

1.1 Describe the research question(s) addressed in this topic

本主题研究市场中的社会责任行为。第一篇文章问，投资者为什么持有社会责任基金：是因为亲社会偏好、社会形象、财务预期，还是风险分散。第二篇文章问，在有价格和竞争的市场交易中，人们是否仍愿意付出成本避免伤害第三方，还是市场会削弱社会责任行为。

This topic studies socially responsible behavior in markets. The first paper asks why investors hold socially responsible mutual funds: because of prosocial preferences, social image, financial expectations, or diversification motives. The second paper asks whether people are still willing to pay costs to avoid harming third parties in competitive markets, or whether market exchange weakens socially responsible behavior.

1.2 Summarize the empirical strategy/design and purpose in this topic

第一篇文章把真实基金持仓、问卷和有激励实验结合起来。研究者用账户数据观察社会责任基金持有，用信任博弈衡量亲社会偏好，用彩票选择衡量风险偏好，用是否经常谈论投资衡量社会形象动机，并用回归比较这些因素的解释力。

第二篇文章设计了实验市场。卖方选择低成本但伤害第三方的产品，或高成本但不伤害第三方的产品；买方观察价格和产品类型后购买。实验还改变竞争、信息和成本条件，并比较市场与非市场选择，目的是检验市场环境如何影响社会责任。

The first paper combines real fund holdings, survey answers, and incentivized experiments. Account data identify SRI holdings, a trust game measures prosocial preferences, lottery choices measure risk attitudes, and talking about investments proxies social image motives. Regressions compare the explanatory power of these factors.

The second paper designs experimental markets. Sellers choose between a cheaper product that harms a third party and a costlier product that avoids harm; buyers observe prices and product types before purchasing. The experiments vary

competition, information, and costs, and compare market with non-market choices to test how markets shape social responsibility.

1.3 Summarize the main empirical/experimental findings in this topic

第一篇文章发现，亲社会偏好和社会形象都能解释社会责任基金持有。更亲社会、也更常谈论投资的人更可能持有这类基金。财务动机有一定作用，但不是主要解释，因为许多社会责任投资者愿意接受较高费用和较低预期收益。

第二篇文章发现，市场没有完全消除社会责任。许多交易选择不伤害第三方的产品，而且这类产品存在价格溢价。但社会责任会受成本和制度环境影响；与非市场选择相比，市场环境会降低社会关切的表达。

The first paper finds that both prosocial preferences and social image explain SRI holdings. Investors who are more prosocial and those who talk more about investments are more likely to hold SRI funds. Financial motives matter, but they are not the main explanation, since many SRI investors accept higher fees and lower expected returns.

The second paper finds that markets do not fully eliminate social responsibility. Many trades involve products that avoid third-party harm, and these products earn a price premium. However, responsible behavior responds to costs and institutional context; compared with non-market choices, market exchange reduces expressed social concern.

1.4 Do you have any critical thoughts on the authors' interpretation of their results? If so, specify them.

一个质疑是，实验环境本身可能改变行为。参与者知道选择被记录，而且第三方损失被明确呈现，这可能使他們更愿意表现得公平或社会负责。因此，实验中的社会责任行为不一定完全等同于真实市场中的自发偏好。

另一个问题是，实验中的第三方完全被动，不能反馈、抵制、惩罚，也不能通过媒体或监管施加压力。现实市场中，这些力量可能影响企业和消费者行为。因此，实验能识别受控条件下的自愿社会责任，但未必完整反映现实市场机制。

One concern is that the experimental setting itself may change behavior. Participants know that their choices are recorded, and third-party harm is made explicit. This may make them more willing to appear fair or socially responsible, so the observed behavior may not fully represent spontaneous preferences in real markets.

Another concern is that third parties in the experiment are passive. They cannot give feedback, boycott, punish, or exert pressure through media or regulation. In real markets, these forces may affect firms and consumers. Thus, the experiment identifies voluntary responsibility under controlled conditions, but may not fully capture real market mechanisms.

1. 说明 Becker-DeGroot-Marschak elicitation method 和 random binary choice method 的方法与优缺点。

Becker-DeGroot-Marschak elicitation method, 简称 BDM, 是让被试报告自己对某个商品、彩票或政策前景的 valuation 或 certainty equivalent, 然后随机抽取一个价格; 如果随机价格低于其报价, 就按随机价格交易, 否则不交易。理论上, 真实报告是最优策略。random binary choice method 则让被试在一系列二元选择中做决定, 例如“确定金额”与“彩票”之间, 或“确定福利合约”与“不确定福利分布”之间, 通过转换点推断其 valuation 或 certainty equivalent。BDM 的优点是 incentive compatible 且信息较连续, 缺点是机制复杂, 被试可能难以理解。random binary choice method 更直观, 理解成本较低, 但通常只能得到一个区间估计, 并可能受到起始值、顺序和锚定影响。

2. 什么是 multiple price list, MPL? 它适合测量什么?

multiple price list 是一种用一系列有序二元选择来推断偏好的方法。常见设计是让被试在“确定金额”和“风险彩票”之间逐行选择, 确定金额逐步增加; 被试从彩票转向确定金额的位置可以作为 risk preference 或 certainty equivalent 的代理变量。它也可以用来测量对某项功能、限制工具或商品的 willingness to pay。优点是形式直观, 便于大样本在线实验和问卷实验; 缺点是测量值依赖选项间距, 只能得到区间或近似转换点, 且若被试多次来回切换, 解释会变得困难。

3. 为什么实验中常使用 random incentive mechanism?

random incentive mechanism 是让被试完成多个选择, 但最终只随机抽取其中一个或少数几个真实支付。它的主要作用是降低实验成本, 同时让每个选择在被试看来都可能影响实际收益, 从而维持 incentive compatibility。它常和 MPL、BDM、风险偏好实验、商品估值实验结合使用。它的潜在问题是被试必须理解“每一题都有可能被抽中”, 否则可能不认真作答; 此外, 如果被试把多项选择看成一个组合, 可能出现跨题目对冲或心理账户问题。

4. 如何用实验测量 certainty equivalent 和 risk premium?

certainty equivalent 是使被试在确定收益和某个风险或不确定前景之间无差异的确定金额。实验中可以直接用 BDM 让被试报价, 也可以用 sequence of binary choices 或 MPL 来逐步 bracket 其确定性等价。在政策不确定性研究中, 研究者先让受访者描述其主观福利分布, 再让其在“确定福利合约”和“主观不确定福利”之间做一系列二元选择; expected value 与 certainty equivalent 的差额就是 risk premium。优点是能估计没有市场价格的政策风险或福利成本, 缺点是问题抽象, 容易受金融素养、概率理解和锚定影响。

5. 如何区分 risk aversion 和 ambiguity aversion?

risk aversion 针对的是概率已知的风险, 例如一个明确的 50-50 彩票; ambiguity aversion 针对的是概率分布本身未知的情形, 例如不知道盒子中红球和黑球的比例。实验上通常将一个已知概率赌局和一个未知概率赌局设计成收益相近或可比, 如果被试系统性偏好已知概率赌局, 就说明存在 ambiguity aversion。这个区分重要, 因为人们可能不是讨厌收益波动本身, 而是讨厌概率不可知。缺点是这类实验对被试的概率理解要求较高, 且有时 ambiguity aversion 会与对实验说明的不信任混在一起。

6. 什么是 compound lottery, 为什么它在 ambiguity 实验中重要?

compound lottery 是多阶段彩票, 例如第一阶段决定盒子构成, 第二阶段再从盒子里抽球。标准期望效用理论通常要求 reduction of compound lotteries, 即多阶段彩票可化简为一个等价的一阶段彩票。ambiguity 实验中引入 compound lottery 的目的, 是判断被试对不确定性的厌恶究竟来自真正的 ambiguity aversion, 还是来自不能或不愿意把复合彩票化简。其优点是能更细致地区分偏好类型; 缺点是设计和理解都较复杂, 回答可能反映计算能力而非纯粹偏好。

7. trust game 如何测量 trust 和 social preferences?

trust game 是两人序贯博弈。第一人决定给第二人多少钱，这笔钱通常会被实验者放大，第二人再决定返还多少。第一人的转账行为通常被解释为 trust，但它也可能混合 altruism、risk preference 和对第二人的信念。第二人的返还行为更常被用于测量 reciprocity 或 intrinsic social preferences，因为完全自利的第二人应返还 0。优点是结构清楚，可以排除重复互动、合约和惩罚等机制；缺点是单一行为仍可能混合多种动机。

8. 什么是 strategy method? 它与 direct-response method 有什么区别?

strategy method 是让被试在不知道对方实际选择之前，对所有可能情形都预先作出条件性反应。例如在 trust game 中，second mover 需要说明在 first mover 可能转来不同金额时分别返还多少。direct-response method 则是等对方实际行动发生后再作出反应。strategy method 的优点是能从同一个被试处获得完整反应函数，提高数据量，也便于测量偏好；缺点是选择更冷静、更抽象，可能弱化真实情境中的情绪反应，因此不一定等同于直接面对他人行为时的即时选择。

9. 如何用实验区分 risk aversion、social preferences 和 betrayal aversion?

一种常见做法是保持收益结构相同，但改变不确定性的来源。若收益由自然状态决定，这是普通 risk；若收益还影响他人，这是 risky dictator game；若收益取决于另一人是否值得信任，这是 trust game。通过比较被试在不同情境中要求的 minimum acceptable probability, MAP，可以判断他们是否对“被人背叛”额外厌恶。如果 trust game 中要求的成功概率高于自然风险或 risky dictator game，说明不只是风险厌恶，而是存在 betrayal aversion。优点是能拆分不同心理来源；缺点是 MAP 问题较抽象，被试可能难以准确表达概率阈值。

10. 如何用实验测量 social image concerns?

测量 social image concerns 的核心是改变行为是否被他人观察，或者改变某一选择能传递的社会信号，同时尽量保持物质收益不变。例如 status goods 实验通过比较“同等功能但无身份外观的产品”和“带身份信号的产品”来识别 status utility；SRI 研究则把真实投资行为、实验测得的社会偏好和“是否经常谈论投资”结合起来，间接测量 social signaling。优点是能把社会形象从单纯消费效用中分离出来；缺点是 social image 往往不能直接观察，代理变量可能混入社交性、自信或金融兴趣。

11. 如何区分 intrinsic preferences 和 signaling motives?

intrinsic preferences 指个体即使在匿名、无人观察时也愿意牺牲金钱来帮助他人或遵守某种价值；signaling motives 指个体行为部分是为了让别人形成某种正面评价。实验上可以通过匿名性、可观察性、是否需要主动披露、以及与真实行为数据的匹配来区分二者。例如 SRI 研究用匿名 one-shot trust game 的 second mover 行为测量 intrinsic social preferences，同时用谈论投资的频率代理 social signaling。优点是能避免把所有亲社会行为都解释为道德偏好；缺点是两种动机在现实中常常同时存在，难以完全分离。

12. 什么是 commitment device? 实验中如何识别其需求?

commitment device 是个体用来约束未来自己的工具，例如屏幕时间限制、自动储蓄、戒烟押金或提前承诺。实验中，如果被试愿意为获得限制工具而放弃金钱，或者自愿设置未来限制并因此改变行为，就可以说明他们预期未来自己可能会失控。digital addiction 实验中，limit treatment 允许被试设置未来生效且不易立即取消的屏幕时间限制，并用 MPL 测量其 valuation。优点是能识别 self-control problems；缺点是使用限制工具也可能来自提醒效应、实验需求效应或单纯好奇。

13. 如何用 temporary incentives 检验 habit formation?

habit formation 指今天的消费会提高未来的消费倾向。实验上可以暂时补贴被试减少某种行为，然后观察补贴

结束后行为是否仍然下降。如果补贴期结束后处理组仍比对照组使用更少，说明过去消费减少改变了未来习惯。**digital addiction** 实验用 **bonus treatment** 暂时奖励减少社交媒体使用，并观察后续使用是否持续降低。优点是识别清楚，直接对应动态消费模型；缺点是持续效应也可能来自学习、提醒或安装监控软件后的行为改变。

14. 为什么实验经常同时收集 **actual behavior**、**beliefs** 和 **predictions**？

actual behavior 告诉我们被试实际做了什么，但不一定能解释原因；**beliefs** 和 **predictions** 能揭示被试对未来、他人行为或风险概率的主观判断。二者结合后，研究者可以区分偏好变化、信念变化、注意力变化和自我控制问题。例如 **digital addiction** 中既记录实际使用，也询问未来使用预测；政策不确定性研究既 **elicit subjective distribution**，也 **elicit certainty equivalent**。优点是机制解释更强；缺点是问预测本身可能改变注意力，且主观回答存在测量误差。

15. **field experiment** 与 **lab experiment** 各有什么优势和弱点？

lab experiment 的优势是控制强，研究者可以规定收益结构、信息条件和互动方式，适合研究基本机制，例如资产泡沫、信任、**ambiguity** 和策略推理；弱点是环境较人工，外部有效性可能有限。**field experiment** 更接近真实决策环境，例如信用卡升级、屏幕时间使用、学生分组或在线购物，外部有效性较强；弱点是环境复杂，处理不一定只改变单一机制，且实施成本较高。二者并非替代关系，较好的设计常用 **lab** 的可控性验证机制，再用 **field setting** 检验现实相关性。

16. 什么是 **induced value**？它在市场实验中为什么重要？

induced value 是研究者在实验中人为规定资产、商品或选择的基本价值，使其对研究者和被试都是明确的。资产泡沫实验尤其依赖 **induced fundamental value**，因为现实市场中很难确定价格是否偏离基本面；实验室则可以规定资产的红利过程或最终价值，再将交易价格与 **fundamental value** 比较。优点是能清楚判断 **mispricing** 和 **bubble**；缺点是被试可能未必完全理解或按照基本价值交易，实验市场也可能过于短期和简化。

17. 为什么市场实验经常使用 **repeated rounds** 或 **experience variation**？

repeated rounds 可以观察学习过程，**experience variation** 可以判断偏离理性预测的行为是否来自缺乏经验。资产泡沫实验中，市场价格是否随着交易者经验增加而接近 **fundamental value**，就是一个重要问题；混合经验市场还可以检验少数无经验者是否足以重新引发泡沫。优点是能研究学习、经验和市场纪律；缺点是重复实验也可能带来策略性适应、疲劳或特定实验规则下的学习，而不一定等同于真实市场经验。

18. 如何用实验研究 **bounded rationality** 和 **strategic thinking**？

研究 **bounded rationality** 的实验通常设计一个有清楚理性预测、但需要多层推理的博弈，然后观察被试是否达到均衡，或他们的选择对应几层推理。**Cognitive Hierarchy model** 允许不同玩家有不同 **sophistication**，**L0** 是朴素反应，**L1** best responds to **L0**，**L2** 再 best responds to **L1**。**bubble game** 也通过 **price cap** 改变排除泡沫所需的 **backward induction** 步数，从而研究推理深度和共同知识问题。优点是能把“非理性”具体化为推理层级或反应噪声；缺点是 **L0** 假设和推理层级可能随情境变化，不一定稳定。

19. **Quantal Response Equilibrium** 和 **Cognitive Hierarchy model** 有什么区别？

Cognitive Hierarchy model 强调玩家有不同推理层级，高层级玩家对低层级玩家作最佳反应；**Quantal Response Equilibrium** 强调玩家可能不是总选最优反应，而是以更高概率选择收益更高的行动，因此行为中存在 **payoff-responsive noise**。前者解释“想了几层”，后者解释“即使知道收益也可能出错”。在一些实验中，两者都能解释偏离 **Nash equilibrium** 的行为，但含义不同。前者更适合解释有限层级推理，后者更适合解释随机错误和策略不确定性。

20. 如何用实验区分偏好效应和信息效应？

一个常见方法是保持实际收益或真实风险不变，只改变被试看见的信息、产品标签或显著性；或者反过来，保持信息展示不变，只改变真实物质收益。**status goods** 实验通过保持信用卡功能不变、只改变是否有身份外观来识别 **status component**；**salience** 类实验则试图寻找“客观风险不变，但风险变得更容易想起”的情境。优点是能把行为变化归因于心理或信息渠道；缺点是很难保证 **treatment** 只改变一个维度，例如标签可能同时改变信任、质量预期和社会信号。

21. 为什么实验设计中需要 **placebo task** 或 **manipulation check**？

placebo task 用来控制“完成一个任务”本身的影响，**manipulation check** 用来确认 **treatment** 是否真的改变了研究者想改变的中间变量。比如 **self-affirmation** 处理若声称提高 **self-esteem**，就需要用量表检验它是否真的提高了 **self-esteem**；对照组也需要做一个形式类似但不应改变 **self-esteem** 的任务。这样可以避免把时间、注意力或写作任务的影响误认为目标机制。缺点是 **manipulation check** 本身可能提醒被试研究目的，从而诱发需求效应。

22. **within-subject design** 与 **between-subject design** 如何权衡？

within-subject design 让同一个人在多个条件下作选择，优点是能控制个体固定差异，提高统计功效；缺点是容易产生顺序效应、学习效应、锚定效应和 **demand effects**。**between-subject design** 让不同被试面对不同条件，优点是减少前后比较带来的污染；缺点是需要更大样本来平衡个体差异。很多实验会结合二者，例如先随机分组，再让部分被试在不同模块中经历不同条件，同时用额外设计检查 **order effects**。

23. 为什么一些 **field experiment** 要主动扩大 **support**，而不只是随机化？

随机化解决的是选择偏误问题，但如果处理变量变化范围太窄，就很难学习不同政策方案的效果，这就是 **support problem**。**peer effects** 研究中，只随机分配但不操纵同伴能力构成，可能只能观察到很小范围内的同伴组合；主动扩大 **peer composition** 的范围后，才可以估计更灵活的 **response function**，并预测不同分组政策的效果。优点是政策外推更有依据；缺点是设计更复杂，且极端组合可能引发额外行为反应。

24. 为什么实验中要关注 **comprehension checks** 和 **attention checks**？

comprehension checks 用来确认被试理解支付规则、概率、税率、限制功能或博弈结构；**attention checks** 用来确认被试认真阅读材料。如果缺少这些检查，异常选择可能只是误解规则，而非真实偏好或认知偏差。尤其是 **BDM**、**MPL**、**certainty equivalent**、**MAP** 和复杂在线购物实验，都要求被试理解机制。缺点是这些检查也可能改变被试注意力，使某些本应不显著的信息变得显著，因此它们提高内部有效性的同时，可能改变自然决策环境。

25. 什么是 **demand effects**？为什么行为实验中特别需要警惕？

demand effects 是指被试猜到实验目的后，按自己认为研究者期待的方式行动。它在社会责任、捐赠、诚实、**self-control**、**status goods** 等主题中尤其重要，因为被试可能想显得公平、诚实、克制或有社会责任感。常见缓解方法包括使用中性语言、匿名决策、真实激励、**placebo task**、隐蔽研究目的，以及让对照组承受类似观察和任务负担。问题在于 **demand effects** 很难完全排除，因此回答实验批判题时可以说，某些处理效应可能不仅反映真实偏好，也可能反映被试对实验意图的反应。

26. 为什么匿名性和可观察性是实验设计中的核心操纵？

匿名性和可观察性能帮助区分“内在偏好”和“形象动机”。如果在匿名条件下仍然牺牲金钱帮助他人，较可能反映 **intrinsic preferences**；如果行为只在他人能观察时显著变化，则说明 **social image** 或 **reputation** 可能重要。信任、**SRI**、**status goods**、**honesty** 和社会责任市场实验都涉及这一逻辑。缺点是实验中的匿名性往往依赖被试相信实

验者真的无法识别其行为，如果被试怀疑匿名性，结果可能仍受形象顾虑影响。

27. 如何把 survey data、experimental data 和 administrative data 结合起来？

综合数据设计通常用 administrative data 观察真实行为，用 survey data 观察信念、动机和背景变量，用 incentivized experiments 测量较干净的偏好参数。例如 SRI 研究把真实基金持仓、问卷中的投资动机和实验中的 social preferences、risk preferences 连接起来。优点是既有真实外部行为，又有机制变量；缺点是调查参与者可能选择性响应，实验测得的偏好也可能不完全等于现实中的长期偏好。

28. 如何用实验研究“机制”而不只是平均处理效应？

研究机制通常需要设计多个处理，使它们分别对应不同理论渠道；也可以收集中间变量、信念、预期或后续行为。例如 digital addiction 中 bonus treatment 更对应 habit formation，limit treatment 更对应 self-control problems；status goods 中功能相同但身份信号不同的产品用于识别 status component；SRI 研究同时区分 financial motives、social preferences 和 signaling motives。优点是解释力强；缺点是机制处理经常不够“纯”，一个 treatment 可能同时改变注意力、信念和偏好。

29. 实验经济学中，什么时候使用 field evidence 比 lab evidence 更有说服力？

当研究对象与真实市场制度、真实消费环境、真实社交关系或长期行为密切相关时，field evidence 通常更有说服力。例如 status goods、digital addiction、peer effects、SRI 和 tax salience 都依赖真实或半真实环境，因为被试的实际消费、投资、学习或手机使用比抽象实验选择更接近研究问题。其不足是 field setting 中干扰因素更多，研究者控制较弱；因此较好的回答应强调 field evidence 提高外部有效性，但不自动保证机制识别更干净。

30. 实验经济学方法的共同优点和共同局限是什么？

共同优点是可以主动创造可识别的 variation，把不可观察的偏好、信念、社会动机或认知偏差转化为可观察选择，并通过激励机制减少空泛问卷回答。共同局限是被试可能误解规则、受到实验者需求效应影响、在低金额环境下表现不同于真实高 stakes 决策，且实验样本和实验场景未必能代表现实市场。较好的考试回答通常不应只说“实验能识别因果”，还应进一步说明识别来自哪里，例如 incentive compatibility、随机化、匿名性、可观察性操纵、induced values、机制处理或真实行为数据匹配。